
USER/DEVELOPER
Bug Submission UI
Bug Description
Stack Trace
Error Logs
File Upload
Backend / API Layer
API Request Handling
File Handling
Request Validation
Bug Report Processing Module
Parsing
Preprocessing
Error Extraction
Bug Report / Submission Database
Submitted Bug Reports
Stack Traces
Error Logs
Metadata
Historical Defect Knowledge Base
Mozilla
Apache
Eclipse
Historical Bug Reports
Past Fixes / Resolutions
Data Cleaning & Chunking Pipeline
Data Cleaning
Text Standardization
Chunking
Embedding Generation Module
Text Embeddings
Semantic Representation
Vector Database / Semantic Search
Embedded Bug Reports
Similarity Search
Metadata
RAG Retrieval Pipeline
Query Embedding
Similarity Retrieval
Relevant Bug Context
LLM
Bug Context Analysis
Reasoning
Fix Generation
AI Agent Layer
Agent Orchestrator
Task Routing
Agent Coordination
Workflow Management
Triage Agent
Log Analysis Agent
Duplicate Detection Agent
Root Cause Agent
Remediation Agent
Structured Findings / Diagnosis
Bug Type
Root Cause
Severity
Duplicate Status
Recommended Fix
Results & Recommendations Interface
Bug Analysis Summary
Root Cause
Recommended Fix
Confidence Score
Final Report
Knowledge Base Update Module
Updated Bug Information
New Resolutions
Validated Fixes